Operating Systems Lab – Pseudocode Collection

Preemptive SJF (SRTF) with Aging

```
BEGIN
INPUT n
INPUT arrival time and burst time
SET completed = 0
SET time = 0
WHILE all processes not completed
SELECT any process whose waiting time > threshold
IF such process exists
choose that process
ELSE
SELECT process with smallest burst time
END IF
IF no process available
increase time
ELSE
execute selected process completely
set completion time
mark as completed
END IF
END WHILE
CALCULATE TAT = CT - AT
CALCULATE WT = TAT - BT
CALCULATE average WT and TAT
CALCULATE throughput
PRINT output
END
```

FCFS Disk Scheduling

```
BEGIN
INPUT n
INPUT request sequence
INPUT initial head position
SET totalSeekTime = 0
FOR each request
calculate distance
add to totalSeekTime
move head
END FOR
PRINT output
END
```

SSTF Disk Scheduling

```
BEGIN
INPUT n
INPUT request sequence
INPUT head
MARK all not visited
FOR each request
find nearest request
move head
update seek time
END FOR
PRINT total seek time
END
```

SCAN Disk Scheduling

```
BEGIN
INPUT requests, head, direction
SORT requests
MOVE in one direction
go to disk end
REVERSE direction
serve remaining
PRINT output
END
```

LOOK Disk Scheduling

```
BEGIN
INPUT requests, head, direction
SORT requests
MOVE till last request
REVERSE direction
serve remaining
PRINT output
END
```

C-SCAN Disk Scheduling

```
BEGIN
INPUT requests, head
SORT requests
MOVE right
go to end
JUMP to start
serve remaining
PRINT output
END
```

C-LOOK Disk Scheduling

```
BEGIN
INPUT requests, head
SORT requests
MOVE right
JUMP to first request
serve remaining
PRINT output
END
```

Dining Philosophers

```
BEGIN
INPUT N
INITIALIZE room = N-1
INITIALIZE forks
CREATE threads

PHILOSOPHER:
WAIT room
WAIT forks
EAT
RELEASE forks
RELEASE room
END
```